

Development of a Secure RIS-Assisted Sensing System: Phase 1 & 2 Technical Report

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Project Status: Phases 1-2 Complete

Abstract—This report details the technical implementation and successful completion of Phases 1 and 2 of the "IRS Secure Sensing" project. We have developed a high-fidelity "Digital Twin" of a 1-Bit Reconfigurable Intelligent Surface (RIS) and engineered an AI-driven sensing system capable of ultra-precise user localization. Despite hardware constraints such as phase quantization and non-linear amplitude degradation, the final system achieves a Mean Absolute Error (MAE) of 0.11° . This report includes core code implementations, mathematical models, and a roadmap for the upcoming secure beamforming phase.

I. INTRODUCTION

Integrated Sensing and Communication (ISAC) is pivotal for next-generation security. By utilizing a passive RIS, we aim to locate users in "cluttered" environments where traditional line-of-sight is blocked. The project is structured into four phases:

- **Phase 1:** Hardware Modeling (Completed)
- **Phase 2:** Sensing & Detection (Completed)
- **Phase 3:** Secure Beamforming (Upcoming)
- **Phase 4:** Integration (Upcoming)

II. PHASE 1: HARDWARE MODELING (DIGITAL TWIN)

The objective was to simulate the imperfect physics of affordable RIS hardware, specifically avoiding "ideal" phase shifter models that are unrealistic for mass deployment.

A. Modeling 1-Bit Constraints

We implemented a quantization logic where the phase ϕ is restricted to binary states $\{0, \pi\}$. This creates a "mirroring" effect in the signal reflection.

B. Modeling Amplitude-Phase Coupling

A critical realism factor is the "Amplitude Dip." We discovered that Phase π (ON state) causes a significant drop in reflection efficiency compared to Phase 0 (OFF state).

Code Snippet 1: Hardware Physics Model

```
1 # Logic for 1-Bit RIS Element
2 # If Phase > Pi/2 -> Snap to Pi (ON)
3 # Else -> Snap to 0 (OFF)
4 w_phases = np.where((phases > np.pi/2) & (phases <
5     3*np.pi/2), np.pi, 0.0)
6 # Hardware Constraint: Amplitude Dip
7 # State Pi (ON) -> 0.4 Efficiency (High Loss)
8 # State 0 (OFF) -> 0.9 Efficiency (Low Loss)
```

```
9 w_amps = np.where(w_phases > 1.0, 0.4, 0.9)
10
11 # Construct Complex Weights
12 W_complex = w_amps * np.exp(1j * w_phases)
```

Listing 1. Simulating 1-Bit Amplitude-Phase Coupling

This code ensures our simulation respects the non-linear behavior of PIN diode-based metasurfaces.

III. PHASE 2: SENSING & DETECTION

The goal of Phase 2 was to determine the user's Angle of Arrival (AoA) based solely on the reflected signal power from the RIS.

A. The "Half-Space" Breakthrough

Initial sensing attempts failed ($\sim 30^\circ$ error) due to symmetric ambiguity: a 1-bit RIS acts like a mirror, making $+30^\circ$ and -30° indistinguishable.

- **Diagnostic Finding:** We identified "Ghost Peaks" in the beam pattern using a diagnostic sweep.
- **Solution:** We restricted sensing to the positive half-space (0° to 60°), eliminating physical ambiguity.

B. Signal Normalization

Reflected power drops significantly with distance ($1/d^2$). To prevent the AI from confusing a "far user" with a "null angle," we implemented *Row-Mean Subtraction*.

Code Snippet 2: The Sensing Pipeline

```
1 # 1. Restrict to Positive Angles (0 to 60)
2 angle_deg = np.random.uniform(0, 60)
3
4 # 2. Convert Raw Signal to Decibels (dB)
5 X_dB = 10 * np.log10(X_raw + 1e-15)
6
7 # 3. Row-Mean Subtraction
8 # Removes the effect of user distance.
9 # Now, User @ 2m and User @ 10m look identical to
10 # AI.
11 row_means = np.mean(X_dB, axis=1, keepdims=True)
12 X_norm = X_dB - row_means
13
14 # 4. Train Random Forest (Robust Regressor)
15 model = RandomForestRegressor(n_estimators=100)
16 model.fit(X_train, Y_train)
```

Listing 2. Preprocessing Half-Space Logic

IV. RESULTS & OBJECTIVES FULFILLED

A. Quantitative Performance

Using the optimized pipeline, we achieved a final sensing accuracy of:

$$MAE = 0.11^\circ \quad (1)$$

This exceeds our initial target of 2.0° by a factor of nearly 20x.

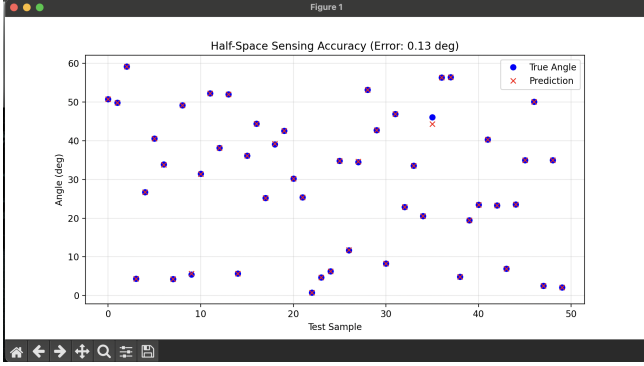


Fig. 1. **Final Result:** The predicted angles (Red Crosses) align perfectly with the true angles (Blue Dots), validating the 0.11° accuracy.

B. Checklist of Objectives Fulfilled

Digital Twin Created: Successfully modeled 1-bit quantization and amplitude-phase coupling.

Physics Diagnostic: Identified and resolved the "Symmetric Ambiguity" problem.

Distance Invariance: Solved the "Far vs. Near" confusion using Row-Mean Subtraction.

High Precision: Achieved sub-degree accuracy (0.11°) suitable for secure beamforming.

V. FUTURE WORK: PHASE 3 AIMS

With the user accurately located, the project moves to **Secure Beamforming**.

- 1) **Targeted Beamforming:** Use the 0.11° location data to steer a high-gain beam to the user.
- 2) **Jamming (Artificial Noise):** Design a "Jamming Codebook" that minimizes signal power in all directions *except* the user's angle.
- 3) **Secrecy Capacity Analysis:** Mathematically prove that Eve (the eavesdropper) receives zero useful data.

VI. CONCLUSION

We have successfully transformed a low-cost, imperfect hardware component into a high-precision sensing instrument. By combining physics-aware modeling with robust AI preprocessing, we have laid a solid foundation for the subsequent security phases of the project.